Universidad de Santander

UDES

VIGILADA MINEDUCACIÓN | SNIES 2832

Data Structures: The Building Blocks of Software

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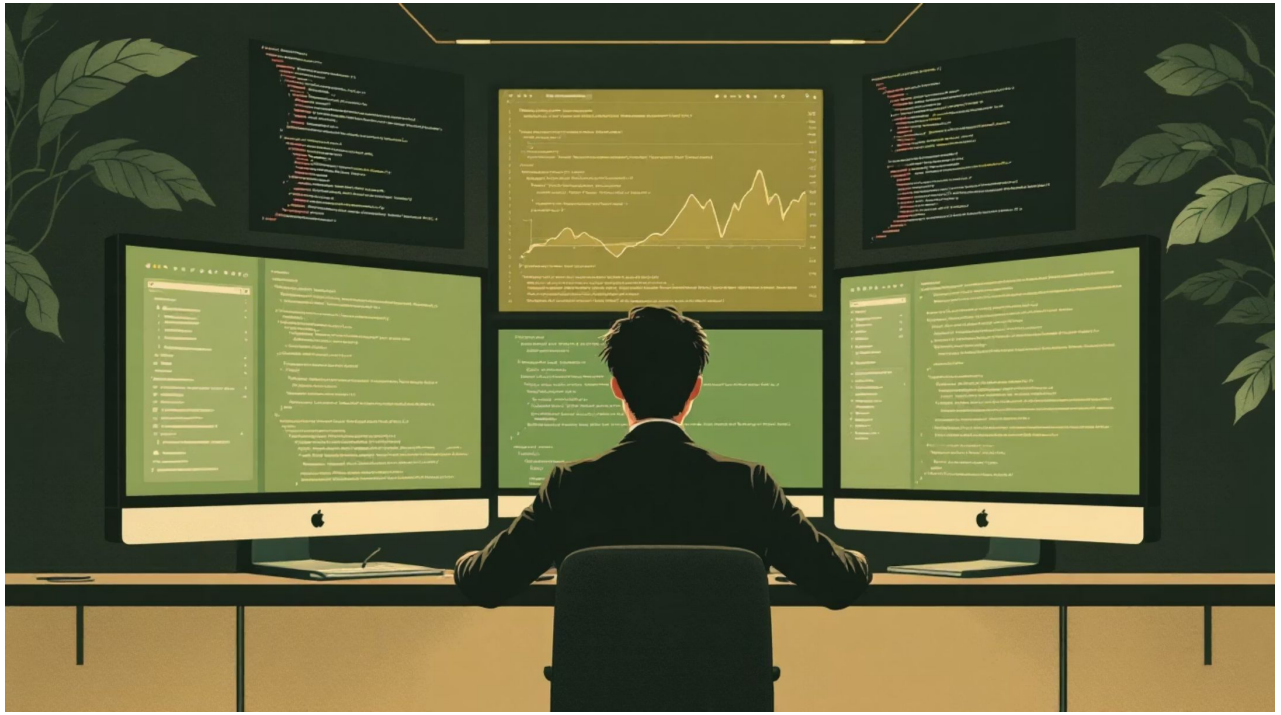
The Critical Importance for Software Development

Efficiency as a Priority

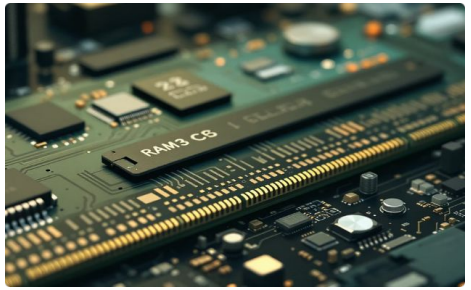
The correct choice of a data structure directly determines the efficiency of any algorithm. A well-selected structure minimizes computational complexity and optimizes system resources. Primitive structures—integers, floats, characters, and pointers—are the foundation directly managed by machine instructions, forming the basis of more complex organizations.



Solving Real-World Computational Problems

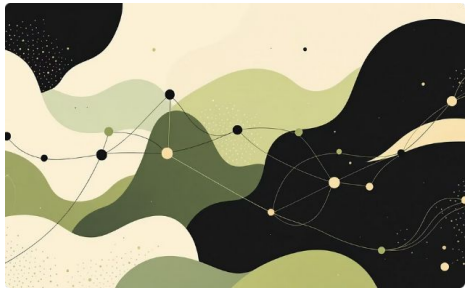
Database Design

Efficient management of large volumes of structured information



Memory Management in OS

Dynamic administration of operating system resources



Network Modeling

Representation of large-scale interconnected systems

Understanding Data Relationships

Pre-Code Analysis

Before coding, we must deeply understand the relationships between data elements to choose the optimal structure.

Types of Relationships

Data can be hierarchical, linear, or interconnected. Each pattern requires a specific structure for efficient representation.

Mathematical Modeling

Data structures are logical models that capture the essence of the problem before implementation.

Queue
Scheduling and data buffering.

Stack
Recursion and 'undo' function.

List
Dynamic memory and links.

Stacks

- Function call management and recursion
- Implementation of 'undo' functionalities

Queues

- Operating system job scheduling
- Data buffering in network communications

Linked Lists

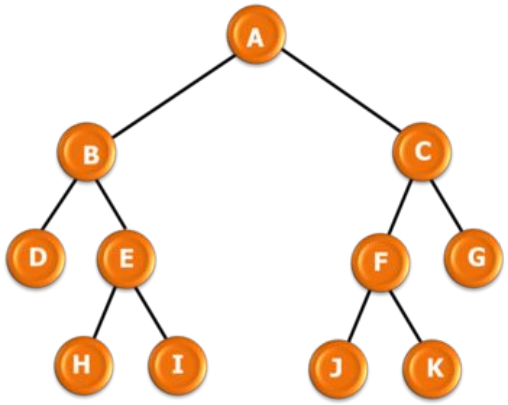
- Dynamic memory management
- Efficient element insertion and deletion

Advanced Structures: Trees and Graphs

Trees

File Systems: Standard representation of directories and files in hierarchical structures.

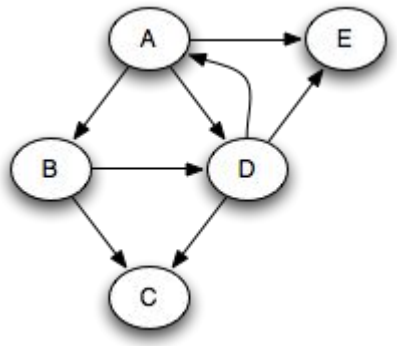
Hierarchical Data: Organization of information with clearly defined parent-child relationships.



Graphs

Social Networks: Modeling complex relationships between users and interconnected communities.

Optimal Paths: Shortest path search algorithms in logistics and GPS navigation systems.



DSA: The Developer's Competitive Edge

Code Scalability

Mastery of Data Structures and Algorithms (DSA) enables writing code that processes massive amounts of data without performance degradation.

Every professional developer must learn DSA to build systems that scale efficiently. This knowledge is fundamental for optimally managing memory and processing power.

The true value lies in developing **algorithmic thinking**: it's not just about making a program work, but making it work as efficiently as possible.

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Professional Engineering Training UDES

01

Algorithmic Thinking

Develop the ability to analyze problems from a computational perspective before implementing solutions

02

Resource Optimization

Learn to efficiently manage memory and processing power in production systems

03

Scalable Architecture

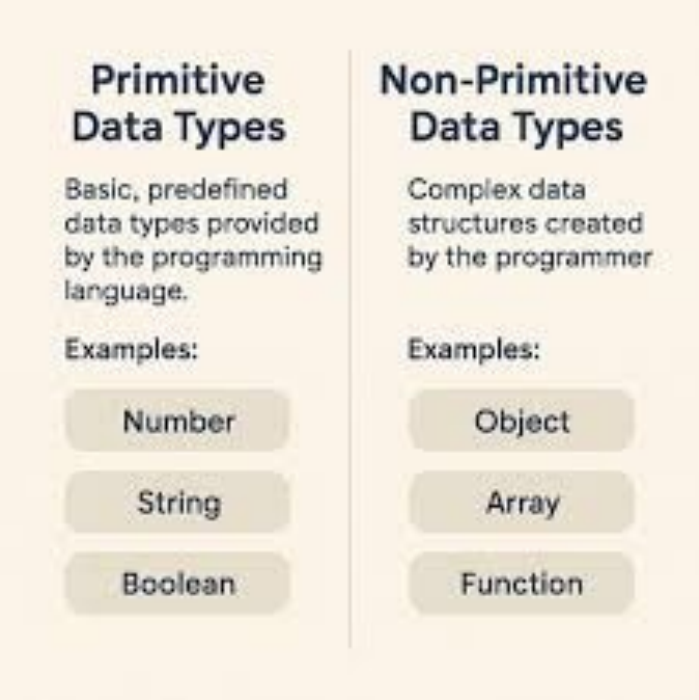
Design systems that grow without compromising performance or stability

From Primitives to Complex Systems

Solid Foundations

Primitive structures —integers, characters, pointers— are handled directly by hardware. Understanding these fundamentals is essential for building more sophisticated abstractions.

Every non-primitive structure is built upon these basic blocks, inheriting their performance characteristics and limitations.



Conclusion: Excellence in Software Engineering

Data structures are much more than technical tools: they are the fundamental language with which we express elegant and efficient computational solutions.

Strategic Choice

The right structure transforms complex problems into manageable solutions

Guaranteed Efficiency

Minimize complexity and optimize system resources

Professional Thinking

Develop the mindset of a world-class software engineer

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GRACIAS